Heat and Thermodynamics - 2

Introduction to Expanding Gas

Basic Principles of Gas Expansion

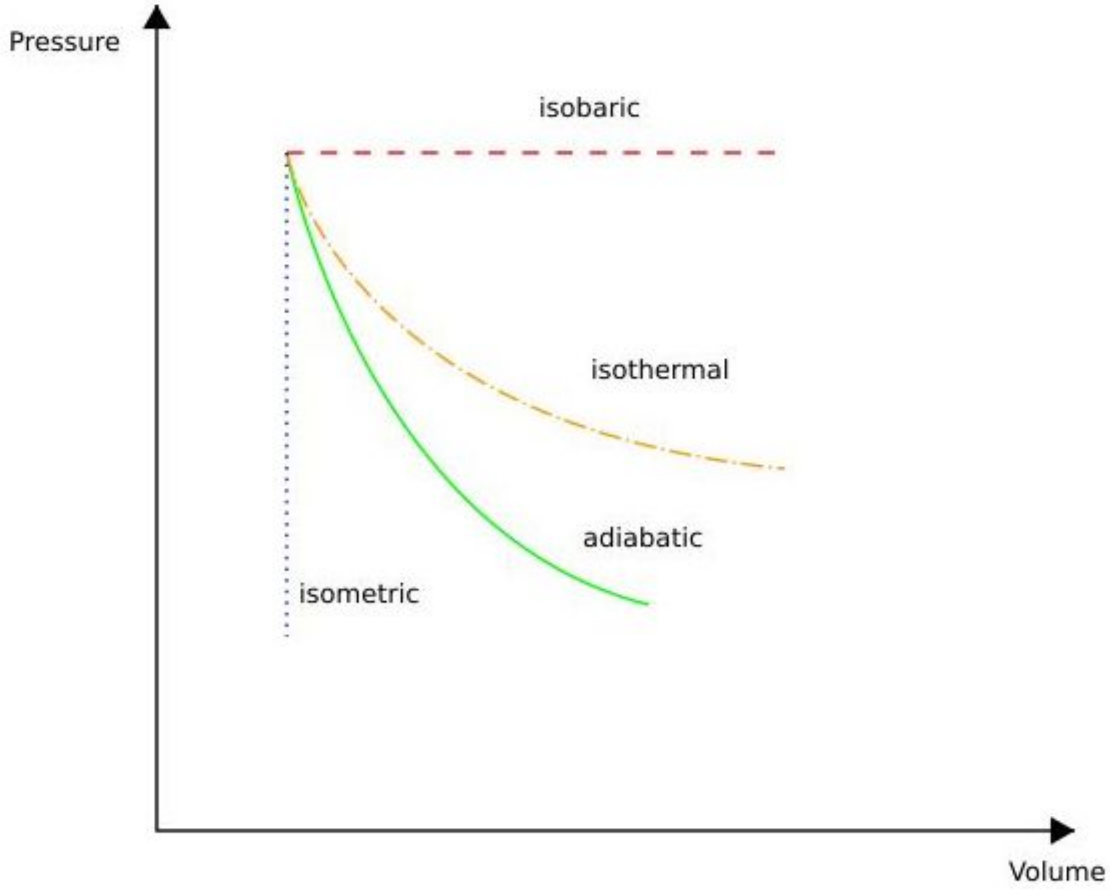
- Ideal Gas Law: $PV=nRT$
- Definition of work in thermodynamics: $W=P\Delta V$
- Types of processes: Isothermal, adiabatic

Isothermal Expansion

- Definition: Expansion at constant temperature.
- Work done in an isothermal process: $W = nRT \ln(Vf/Vi)$
- Example: Isothermal expansion in a piston.

Adiabatic Expansion

- Definition: Expansion without heat exchange.
- Adiabatic process equation: $PV^\gamma = \text{constant}$
- Work done in adiabatic process: $W = (P_i V_i - P_f V_f)/(\gamma - 1)$
- Example: Rapid compression/expansion in engines.


Pressure
isobaric
isothermal
adiabatic
isometric
Volume

Elasticity of a Perfect Gas

Elasticity of a Perfect Gas

- Definition of Elasticity: Elasticity refers to the ability of a material to return to its original shape after being deformed. For gases, it measures the response of gas pressure to changes in volume.
- Perfect (Ideal) Gas: A theoretical gas composed of many randomly moving point particles that interact only through elastic collisions and adhere to the ideal gas law $PV=nRT$
- Importance: Understanding gas elasticity is crucial in fields like engineering, meteorology, and various industrial applications where gas compression and expansion occur.

Basic Principles

- Ideal Gas Law: $PV=nRT$ where
  - P is pressure,
  - V is volume,
  - n is the number of moles,
  - R is the gas constant, and
  - T is temperature.
- Bulk Modulus (K): A measure of a material's resistance to uniform compression, defined as $K=-V(\partial P/\partial V)$
- Relation to Elasticity: For gases, the bulk modulus indicates how the pressure changes with volume under different conditions (isothermal or adiabatic).

Understanding Bulk Modulus

- Physical Meaning: Bulk modulus K quantifies a gas's resistance to compression.
- A higher K means the gas is less compressible.
- Units: The bulk modulus is measured in Pascals (Pa), which is the same unit as pressure.
- Calculation: $K=-V(\partial P/\partial V)$
    - For a small volume change ΔV, the corresponding pressure change ΔP is given by
    - $K=-V(\Delta P/\Delta V)$

Isothermal Elasticity

- Definition: Elasticity measured at constant temperature.
- Isothermal Bulk Modulus: $KT=P$ where P is the pressure of the gas.
- Explanation: Under constant temperature, any change in volume results in a proportional change in pressure, making the bulk modulus equal to the gas pressure.
- Example: Consider a gas confined in a cylinder with a movable piston. When the piston is pushed or pulled slowly, allowing heat exchange with the surroundings, the process is isothermal.

Adiabatic Elasticity

- Definition: Elasticity measured without any heat exchange with the surroundings.
- Adiabatic Bulk Modulus: $KS=\gamma P$ where γ is the adiabatic index (ratio of specific heats C_p/C_v P is the pressure.
- Explanation: In an adiabatic process, the pressure-volume relationship follows $PV\gamma=\text{constant}$. Since no heat is transferred, the pressure change is greater for a given volume change compared to isothermal conditions.
- Example: Rapid compression or expansion of a gas in an insulated cylinder, such as in internal combustion engines.

Applications and Importance

- Engineering: Design of pressure vessels, pneumatic systems, and understanding the behavior of gases in engines and turbines.
- Meteorology: Prediction of atmospheric pressure changes, weather patterns, and understanding the behavior of air masses.
- Industrial Processes: Applications in processes involving gas compression and expansion, such as refrigeration, air conditioning, and manufacturing.
- Conclusion: Understanding the elasticity of gases under different conditions (isothermal and adiabatic) helps in designing efficient systems and predicting gas behavior in various real-life scenarios.

Laws of Thermodynamics

Zeroth Law of Thermodynamics:

- Statement: If two systems are in thermal equilibrium with a third system, they are in thermal equilibrium with each other.
- Significance: This law establishes the concept of temperature and provides the basis for the measurement and comparison of temperatures.

First Law of Thermodynamics (Conservation of Energy)

- Statement: Energy cannot be created or destroyed in an isolated system; it can only change forms.
- Significance: This law is a restatement of the principle of conservation of energy, which states that the total energy of an isolated system remains constant over time. It forms the basis for understanding energy transfer, work, and heat in various processes.

Second Law of Thermodynamics

- Statement: In any energy exchange, if no energy enters or leaves the system, the potential energy of the state will always be less than that of the initial state. Or, in simpler terms, heat cannot spontaneously flow from a colder location to a hotter location.
- Significance: This law defines the direction of natural processes, indicating that spontaneous processes tend to increase the entropy (disorder) of a system. It is essential for understanding energy efficiency, irreversibility, and the limitations of heat engines and refrigeration systems.

Third Law of Thermodynamics:

- Statement: As temperature approaches absolute zero (0 Kelvin or -273.15 degrees Celsius), the entropy of a perfect crystal approaches a minimum value.
- Significance: This law establishes the behavior of entropy at absolute zero temperature and provides a basis for calculating absolute entropies. It is particularly relevant in the study of quantum mechanics and low-temperature physics.

Introduction to Carnot's Cycle

Understanding Carnot's Cycle

- Definition: Carnot's Cycle is a theoretical thermodynamic cycle that describes the most efficient heat engine possible, operating between two temperature reservoirs.
- Concept: It consists of four reversible processes: isothermal expansion, adiabatic expansion, isothermal compression, and adiabatic compression.
- Importance: Carnot's Cycle provides a theoretical upper limit for the efficiency of heat engines and serves as a benchmark for real-world engine performance.

Carnot's Theorem

No real heat engine operating between two energy reservoirs can be more efficient than a Carnot engine operating between the same two reservoirs.

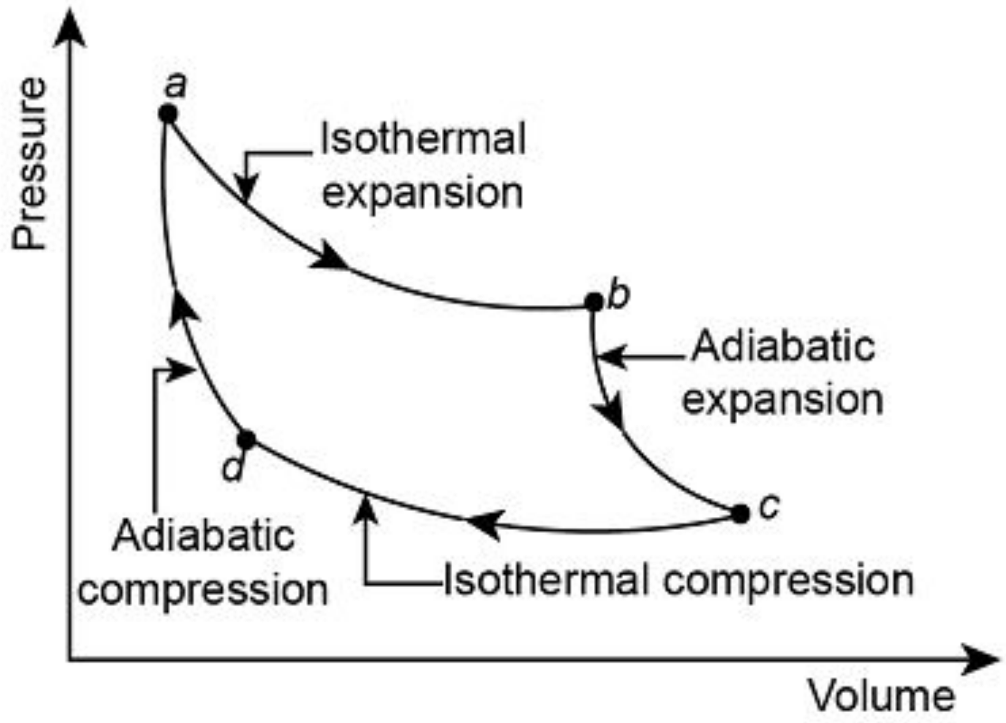
Four Stages of Carnot's Cycle

1. Isothermal Expansion (1-2): Heat is absorbed from the high-temperature reservoir at constant temperature (T_H), causing the gas to expand and do work on the surroundings.

2. Adiabatic Expansion (2-3): The gas continues to expand adiabatically (without heat exchange) and cools down to a lower temperature (T_C).

3. Isothermal Compression (3-4): The gas is compressed at constant low temperature (T_C), and work is done on the gas, releasing heat to the low-temperature reservoir.

4. Adiabatic Compression (4-1): The gas is compressed adiabatically, raising its temperature back to the initial high-temperature (T_H).



Pressure
a
Isothermal expansion
b
Adiabatic expansion
c
d
Adiabatic compression
Isothermal compression
Volume